

6(a)	$\frac{\text{work (done) / energy (transferred from electrical to other forms)}}{\text{charge}}$	B1
6(b)	$R = \rho L / A$	B1
	$V = LA$ and (so) $R = \rho V / A^2$ (with ρ and V constant)	B1
6(c)	$E = IR + Ir$ or $E = I(R + r)$ or $E - Ir = IR$ and $R = (E / I) - r$	A1
6(d)(i)	$P = I^2 R$ or $P = IV$ or $P = V^2 / R$	C1
	$R = 5.4 (\Omega)$ or $V = 10.8 (\text{V})$	C1
	$P = 2.0^2 \times 5.4$ $= 22 \text{ W}$	A1
6(d)(ii)	1. $r = 0.60 \Omega$	A1
	2. $E = \text{gradient}$	C1
	$= \text{e.g. } 5.4 / 0.45$ $= 12 \text{ V}$	A1